

DTSA 5515

Fair Machine Learning: Foundations

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Elective	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective	ML Engineer (69.2 / 100)	60.6 / 100
SPECIALIZATION Fair Machine Learning		

61

CURRENT AVG SCORE
across 6 job roles

96

POTENTIAL SCORE
with recommended additions

+46

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

This course will introduce students to core ideas in machine learning fairness, including philosophical foundations, basic definitions and distinctions, and on-going debates about how fairness can be defined and operationalized in technical systems. We will define different types of bias and how they are introduced into ML systems. Students will learn about fairness requirements under various regulatory regimes including the GDPR, and types of audits that can be performed to study fairness in existing systems.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
ML Engineer	69.2	99.7	+30.5	HIGH
AI Engineer	66.9	100.0	+33.1	HIGH
Machine Learning Engineer	58.7	89.8	+31.1	HIGH
Applied Scientist	58.5	90.7	+32.3	HIGH
MLOps Engineer	56.9	94.9	+38.0	HIGH
Data Scientist	53.5	100.0	+46.5	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

Deep Learning

1 of 6 roles

cnn

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1

LOW

Optional: Deep Learning — would broaden appeal to one additional role. Candidate terms: cnn.

GAP KEYWORDS BY ROLE

ML Engineer	69.2 → 99.7 (+30.5 pts)
cnn media data models model world build problems	
AI Engineer	66.9 → 100.0 (+33.1 pts)
following data models build model problems detection information	
Machine Learning Engineer	58.7 → 89.8 (+31.1 pts)
data interfaces web sets tools design science models	
Applied Scientist	58.5 → 90.7 (+32.3 pts)
research problems models data industry design world science	
MLOps Engineer	56.9 → 94.9 (+38.0 pts)
detection data model text media image tools models	
Data Scientist	53.5 → 100.0 (+46.5 pts)
data sets product mitigation behaviors social build people	

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum–Industry Alignment project team.